

HNSS08 · Brain Development — Stages Chart

The stages of brain development in order, what drives each one, and what happens when it fails | **STEP 1** = beyond class content · **EXAM EMPHASIS** = flagged in class as testable

HoyaDocs
Updated Oct 1, 2026 · Rev 1

Prenatal Stages of Brain Development (1–6)

Order

Induction → Neurulation → Vesicles → Migration → Axons → Competition → Synapses → Myelin → Pruning → Expansion → Aging

Hook

"I Never Vote Means A Citizen Skips My Political Engagement Always"

STAGE	WHEN	WHAT HAPPENS	KEY SIGNALS / CELLS	IF IT FAILS → OUTCOME
1 · Neural induction <i>ectoderm chooses a neural fate</i>	Earliest embryo	Embryonic ectoderm differentiates to a neural fate because of external molecular signals/gradients. The organizer region secretes BMP inhibitors → nearby cells become neurons + glia; cells exposed to BMP become epidermis. Induction = differentiation and loss of potential	BMP (bone morphogenetic protein) vs BMP inhibitors (e.g., Noggin)	Natural (spontaneous) abortion — not seen in clinic. Extra organizer → second body axis → conjoined twins
2 · Neurulation <i>the plate folds into a tube</i>	Weeks 3–4 of gestation	The neural plate folds and forms the neural tube , zipping closed from the middle outward. Rostral tube → brain; caudal → spinal cord. Signaling centers pattern it	Roof plate: BMPs, retinoic acid · Floor plate: SHH (key inducer) · Somites: BMPs	Neural tube defects: anencephaly, encephalocele, craniorachischisis, iniencephaly, open/closed spina bifida (below)
3 · Brain divisions <i>3 vesicles, then 5</i>	5 vesicles by gestational week 6	Regions of the rostral tube grow at different rates , fold and twist: 3 vesicles (prosencephalon, mesencephalon, rhombencephalon) → 5 vesicles (telencephalon, diencephalon, mesencephalon, metencephalon, myelencephalon). Optic vesicles grow out; the tube is filled with CSF	Differential growth (telencephalon expands into the whole cortex)	Failed forebrain splitting → holoprosencephaly (SHH, trisomy 13, FAS) STEP 1
4 · Proliferation + migration <i>building the 6-layer cortex</i>	Gestational weeks 7–18	Progenitors in the ventricular zone divide symmetrically (→ 2 progenitors), then asymmetrically (→ 1 progenitor + 1 non-dividing neuron). Excitatory neurons climb radial glia to the marginal zone , layers built inside-out (deepest first). Inhibitory neurons migrate tangentially . Neocortex = 6 layers	DCX (doublecortin): keeps neurons moving along radial glia · Reelin : tells neurons when to stop and detach	Lissencephaly (smooth brain): abortion, seizures, early death (severe ~10 yr life expectancy). Reelin (recessive) → disorganized layers; DCX → double cortex (laminar heterotopia)
5 · Axon generation + guidance <i>wiring the circuits</i>	Prenatal	Neurons become polar : neurites form, one becomes the axon , the rest dendrites (apical vs basal markers). The axon's motile growth cone senses its environment and is attracted or repelled by diffuse gradients and non-diffuse extracellular signals	Eph receptors + ephrins . Optic chiasm: radial glia express ephrin-B2 ; temporal axons with EphB1 are repelled (stay ipsilateral); nasal axons cross	Aggenesis of the corpus callosum (DCC gene mutations; also FAS): diagnosed in the first 2 yr; intellectual disability, psychosocial symptoms, or none; a seizure may be the first sign
6 · Neuronal competition <i>survival of the connected</i>	Prenatal	Neurons are overproduced . Neurotrophic factor hypothesis : targets secrete limited neurotrophic factor; neurons that don't get enough undergo apoptosis	Neurotrophins (NGF, BDNF, NT-3, NT-4/5), BMPs, hedgehogs; TGF-β ; interleukins	—

Neural tube defects (neurulation failures)

DEFECT	WHAT IT IS
Anencephaly	Much of the brain (cranial tube) fails to form
Encephalocele	Brain tissue protrudes through the skull
Craniorachischisis	Both brain and spinal cord stay open
Iniencephaly	Severe defect at the back of the head and neck (shown in lecture)
Spina bifida (open / closed)	Lower (caudal) tube fails to close

NTD risk factors

Risk factors: low **folate / folic acid**, maternal **insulin-dependent diabetes**, certain maternal **anticonvulsants**; excess **vitamin A / isotretinoin** (retinoic acid is a roof-plate signal). Folic acid fortification cut cases, yet ~**3,000 US pregnancies** a year are affected; rates are higher in **Hispanic** women. ↑ AFP + acetylcholinesterase **STEP 1**.

★ **USMLE high-yield:** Induction (block BMP; failure → abortion) → **neurulation** (wk 3–4; floor plate **SHH**, roof plate **BMP + RA**; failure → NTDs; folate) → **5 vesicles** by wk 6 → **migration** (wk 7–18, ventricular zone, radial glia, **inside-out**; **DCX** = go, **Reelin** = stop; failure → **lissencephaly**) → **axon guidance** (growth cone, Eph/ephrin; **DCC** → callosal agenesis) → **competition** (neurotrophic factors; losers apoptose).

Postnatal Stages (7–11) & Maturation Timeline

STAGE	WHEN	WHAT HAPPENS	KEY SIGNALS / CELLS	IF IT FAILS → OUTCOME
7 · Synaptogenesis <i>making connections</i>	Before birth → throughout life	Pre- and postsynaptic elements regulate each other ; synapses form in steps; glia promote them. Postnatal growth in cortical volume = dendritic growth, spine formation, synaptogenesis — not new neurons	Glial signals; dendritic spines	Abnormal spine morphology ↔ intellectual disability: Fragile X → ↑ spine density ; Rett → ↓ spine density
8 · Myelination <i>speeding conduction</i>	Late fetal → childhood	Myelin drastically improves conduction speed ; extent of myelination reflects stabilization and maturation of axons	Oligodendrocytes (CNS), interacting with signals from each axon (Schwann cells in PNS)	—
9 · Synaptic pruning <i>keeping what works</i>	Childhood → adolescence	Not all synapses are kept; pruning forms functional, specific circuits (like neuronal competition). Some changes depend on experience (plasticity lecture)	Experience / external stimulation	—
10 · Gray + white matter expansion <i>the long maturation</i>	Birth → early 20s	Rapid cortical volume increase in the first 3 postnatal years , then levels off and declines. Regions mature at different rates (table below) → different windows of sensitivity to environment	Myelination + pruning	—
11 · Healthy aging <i>gradual change</i>	Late adulthood	Gray matter loss (larger ventricles + sulci), ↓ synaptic density ; some cognitive and sensory processes decline (working memory after the 7th decade; vocabulary stable)	—	Only some develop severe degeneration: Alzheimer, other dementias, vascular disease

Cortical maturation by age

AGE	WHAT HAS MATURED
4 yr	Primary somatosensory, visual and motor cortex fully developed
6 yr	Language in early stages; prefrontal cortex very immature
9 yr	Parietal lobes and basal ganglia beginning to mature
11 yr	Limbic system, parietal lobe and prefrontal cortex increasing, not yet mature
15 yr	Association cortex maturing through efficient pruning
18 yr	Many more association areas (e.g., prefrontal cortex) reaching maturity
21 yr	Most regions adult — prefrontal cortex and superior temporal lobe still maturing

Rules for every stage

Location-based signals

development runs on signaling pathways; genes or teratogens that disrupt them cause defects

Reused molecules

one family (e.g., BMPs) acts at many stages → deficits can be broad

Timing decides

the process active at the time of the hit is the one that breaks

Earlier failure = worse outcome

Induction → abortion → Neurulation → NTD → Migration → lissencephaly →

Guidance → callosal agenesis → Synapses → intellectual disability

★ **USMLE high-yield:** After birth, neuron number is basically set: brain growth comes from **dendrites, spines and synapses**, then **myelination** and **pruning**. Primary sensory/motor cortex matures first (~age 4); **prefrontal cortex last** (still maturing past 21). Spine density: **Fragile X** ↑, **Rett** ↓. Healthy aging = gray matter loss, bigger ventricles and sulci, fewer synapses.

Environmental Cues, Teratogens & Genes: Timing Is Everything

Fetal alcohol syndrome depends on when the exposure happens

HEAVY DRINKING IN...	DEVELOPING PROCESSES HIT	OUTCOME
1st trimester	Neurons develop, multiply, begin migrating	Facial dysmorphologies (smooth philtrum, thin upper lip, short palpebral fissures), musculoskeletal anomalies, CNS dysfunction , very low IQ
2nd trimester	Migration, branching, synapse formation	Disrupted learning, memory, cognition, executive function; emotional dysfunction
3rd trimester	Synapses, pruning, myelination	Poor coordination , timing difficulties, impaired cognition and memory, emotional dysfunction

Genes and infections by stage

CAUSE	HITS STAGE	RESULT
Reelin (recessive), DCX	4 · migration	Lissencephaly; laminar heterotopia
DCC	5 · axon guidance	Agenesis of the corpus callosum
Fragile X / Rett	7 · synaptogenesis	↑ / ↓ spine density; intellectual disability
CMV	Multiple	Most common infectious cause of birth defects in the US : ~1 in 200 born infected, 1 in 5 of those with problems — hearing loss , developmental/motor delay, microcephaly
Rubella, Zika	Multiple	Brain malformation (Zika: severe microcephaly STEP 1)

★ **USMLE high-yield**: A **teratogen** = an agent that causes abnormality after fetal exposure in pregnancy. Which stage is running decides the defect: **1st-trimester alcohol** → face + very low IQ; **3rd-trimester alcohol** → coordination, memory, emotion. **CMV** = most common infectious cause of birth defects in the US (hearing loss, microcephaly, delay; periventricular calcifications **STEP 1**).